

■ ARDUINO & BLUETOOTH (HC-05)

Smartphone Control — Lesson Notes

■ Topic Arduino + HC-05 Bluetooth Smartphone Control	■ Level Grade 6 Technology / Computing	■ Duration 60 – 90 minutes Hands-on session
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1. What is Arduino?

Arduino is a small, friendly **microcontroller board** that lets you control electronic components by writing simple code. You tell it what to do — and it follows your instructions!

■ Control LEDs ■ Move Robots	■ Spin Motors ■ Make Games	■ Read Sensors ■ Wireless Control
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2. What is Bluetooth?

Bluetooth is a **short-range wireless technology** that lets devices talk to each other without cables. Your headphones, speakers, and keyboard all use it every day!

■ Range	~10 metres (33 ft)
■ Speed	Up to 3 Mbps (Classic BT)
■ Power	Very low power consumption
■ Fun Fact	Named after King Harald Bluetooth of Denmark — who united his people, just like Bluetooth unites our devices!

3. Meet the HC-05 Bluetooth Module

The HC-05 is a compact Bluetooth module that plugs into your Arduino board. It acts as a **wireless bridge** between your phone and your project.

Pin	Connects to	Purpose
VCC	Arduino 5V	Power supply for the module
GND	Arduino GND	Common ground (completes circuit)
TX	Arduino RX (Pin 0)	Transmits data OUT from HC-05
RX	Arduino TX (Pin 1)	Receives data INTO HC-05
LED	(built-in)	Status indicator — blinks when searching, solid when paired

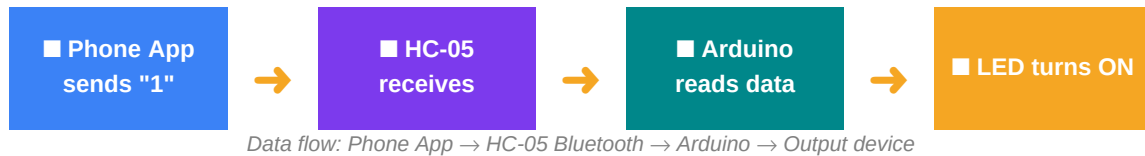
Key Specifications

- Wireless range: approximately 10 metres
- Uses serial (UART) communication protocol
- Default baud rate: 9600

- Default pairing password: **1234**
- Operating voltage: 3.3 V – 5 V

4. How Bluetooth Communication Works

When you send a command from your phone app, the data travels as **serial signals** through the air to the HC-05 module, which passes it along to the Arduino.



5. Pairing HC-05 with Your Smartphone

Step 1	Power on the Arduino + HC-05 circuit. LED on HC-05 will blink rapidly.
Step 2	On your phone, open Settings → Bluetooth and turn it ON.
Step 3	Scan for devices — you will see "HC-05" in the list.
Step 4	Tap "HC-05" and enter the pairing password: 1234
Step 5	LED will now blink slowly (about once every 2 seconds). You are connected! ■

6. Writing Basic Arduino Bluetooth Code

Arduino uses two key functions for Bluetooth communication:

Function	What it does
<code>Serial.begin(9600)</code>	Starts serial communication at 9600 baud rate (set in setup())
<code>Serial.available()</code>	Checks if data has arrived from the phone
<code>Serial.read()</code>	Reads one character of incoming data from HC-05
<code>Serial.print()</code>	Sends data back to the phone app

Sample Sketch — LED Control via Bluetooth

```

int LED = 13;
char cmd;

void setup() {
  Serial.begin(9600); // start Bluetooth comm
  pinMode(LED, OUTPUT);
}

void loop() {
  if (Serial.available()) {

```

```
cmd = Serial.read(); // read 1 char from phone
if (cmd == '1') digitalWrite(LED, HIGH); // LED ON
if (cmd == '0') digitalWrite(LED, LOW); // LED OFF
}
}
```

7. Hands-On Projects

■ Project 1 LED Control	Arduino, HC-05, LED, 220Ω resistor, jumper wires	Send '1' → LED ON Send '0' → LED OFF
■ Project 2 Motor Control	Arduino, HC-05, DC motor, L298N motor driver, wires	'F' Forward 'B' Backward 'S' Stop

8. Safety & Best Practices

■ Disconnect Power	Always turn OFF and unplug your Arduino BEFORE connecting or changing any wires.
■ Use Resistors	Always place a 220Ω resistor in series with LEDs to prevent burnout.
■ Keep Water Away	Never allow liquids near your electronics. Water + electricity = danger!
■ Double-Check Wiring	Before powering on, verify all pin connections match the wiring diagram.

9. Troubleshooting Common Problems

■ HC-05 LED blinks fast / not pairing	Check that VCC is connected to 5V and GND is connected. Try power-cycling the module.
■ Arduino not responding to commands	Verify TX→RX and RX→TX cross-wiring. Re-upload your sketch. Check baud rate is 9600.
■ LED / Motor not working	Check all connections. Ensure resistor is in place. Confirm pin numbers match your code.
■ Cannot find HC-05 in Bluetooth scan	HC-05 must be powered (red/blue LED blinking). Stay within 10 m of the module.
■ How to reset HC-05	Hold the small button on HC-05 while powering on to enter AT command mode.

10. Fun Bluetooth Project Ideas ■

Project	What you build
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■ Remote-Controlled Car	Use F/B/L/R commands to drive a robot car from your phone!
■ Wireless Home Lighting	Control multiple LEDs or a relay to switch real lights.
■ Bluetooth Robot Arm	Send angle values to servo motors to move a robot arm.
■ Smart Fan Control	Adjust a DC fan speed with a slider in your phone app.
■ Temperature Monitor	Arduino reads a sensor and sends temperature data to phone.
■ Musical LED Display	Phone sends BPM values — LEDs flash in sync with music.

11. Summary — What You Learned Today

■ Arduino	A programmable mini-computer board that controls electronics via code.
■ Bluetooth	Short-range wireless tech (≈ 10 m) for device-to-device communication.
■ HC-05	Bluetooth module (VCC, GND, TX, RX, LED) — plugs into Arduino.
■ Pairing	Power HC-05 → scan on phone → select HC-05 → enter 1234 → connected.
■ Code	Serial.begin(9600) in setup(); Serial.read() in loop() to get commands.
■ Safety	Always disconnect power before wiring; use resistors; keep water away.

■ Keep Building Amazing Things!